

AERIAL EYE TO GUARD AGAINST TERRORISM

An eye in the sky to monitor and guide anti-terrorism campaigns—that's the power given to defend us by a new unmanned aerial vehicle developed by a startup company incubated at IIT Bombay



The completely autonomous hovering unmanned aerial vehicle named Netra promises to be of great help to counter insurgency operations

■ VANISHA JOSEPH

The thought of terrorist attacks in Mumbai on Nov. 26, 2008 still sends chills down our spines. Be it the attack on a city that never sleeps or the daily battle fought by our soldiers on snow-capped mountains at the line of control (LoC), every terror attack throws up harder challenges for the armed forces, making the urge for a watchful eye stronger.

Netra, the unmanned aerial vehicle (UAV), which is like a small eye zooming in the sky, can gather real-time intelligence on the ground and become

the perfect weapon to outsmart terror elements. Developed by ideaForge, a start-up company incubated at IIT Bombay, it is a completely autonomous system controlled by a ground-control software application with an intuitive graphical user interface.

"Netra is a completely autonomous system for surveillance and reconnaissance requiring no prior flying experience for an operator. The operator just needs to point and click on the satellite map displayed in the software application for navigation. It keeps sending live video to the ground control station, helping the

users (defence personnel, etc) plan their moves based on the aerial video data received," says Ankit Mehta, managing director, ideaForge. It can help in monitoring counter-terrorist operations, insurgency in forested areas, hostage situations and border infiltration besides conducting local law enforcement operations, search-and-rescue operations, disaster management, etc.

Eye in the Sky!

With many UAVs developed and tested in the past to address our defence needs, what makes Netra a cut above the rest? Netra outsmarts other UAVs in the market with features like vertical take-off and landing, which result in greater operability, BlueFire Autopilot, which is the smallest and lightest autopilot, and fail-safe modes, wherein the UAV automatically returns to the base in case of low battery or communication loss.

High operability. Netra has set new yardsticks in terms of operability. It can vertically take off and land, allowing it to be operated from confined areas and even rooftops.

"The on-board GPS keeps updating the current position of the vehicle. It stores the take-off GPS position that can act as the landing spot too and be changed in flight, if required. Once it is commanded to land, it decides its path and reaches the landing spot. Then it lands vertically down," says Mehta.

Netra is well-equipped to with-